Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<think>
The de Bro gl ie wavelength of a particle is given by Î» = h / p , where Î» is the wavelength , h is Plan ck 's constant , and p is the momentum of the particle . The kinetic energy of the electron is given as 1 e V , which is equal to 1 . 6 ÃĹ 10 ^- 19 J . We can calculate the momentum of the electron using the formula p = âĪļ (2 m K), where m is the mass of the electron and K is the kinetic energy .

First , we need to convert the kinetic energy from e V to Jou les : 1 e V = 1 . 6 ÃĹ 10 ^- 19 J .

Now , we can calculate the momentum of the electron : p = âĪļ (2 ÃĹ 9 . 11 ÃĹ 10 ^- 31 kg ÃĹ 1 . 6 ÃĹ 10 ^- 19 J) = 2 . 26 ÃĹ 10 ^- 24 kg m / s .

Now , we can calculate the de Bro gl ie wavelength : Î» = h / p = 6 . 626 ÃĹ 10 ^- 34 J s / (2 . 26 ÃĹ 10 ^- 24 kg m / s) = 2 . 92 ÃĹ 10 ^- 10 m = 2 . 92 ÃĹ 10 ^- 10 m .

To convert this to cent imeters , we multiply by 10 ^ 2 : Î» = 2 . 92 ÃĹ 10 ^- 8 cm .

However , this value is not in the answer choices . Let 's revisit the calculation .

The de Bro gl ie wavelength is given by Î» = h / p = h / m v , where m is the mass of the electron .

The mass of an electron is 9 . 11 ÃĹ 10 ^- 31 kg . We can calculate the momentum using the kinetic energy :

p = âĪļ (2 m K) = âĪļ (2 ÃĹ 9 . 11 ÃĹ 10 ^- 31 kg ÃĹ 1 . 6 ÃĹ 10 ^- 19 J) = 2 . 26 ÃĹ 10 ^- 24 kg m / s .

Now , we can calculate the de Bro gl ie wavelength :

Î» = h / p = 6 . 626 ÃĹ 10 ^- 34 J s / (2 . 26 ÃĹ 10 ^- 24 kg m / s) = 2 . 92 ÃĹ 10 ^- 10 m = 0 . 292 ÃĹ 10 ^- 8 m = 2 . 92 ÃĹ 10 ^- 8 cm

However , this value is not in the answer choices .